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GROUP C PROJECT

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MODEL COMPARISON

Table 1: Model comparison of all the models in group

Sr#	MODEL NAME	MODEL STATISTICS			Feature Selected	Distance Matrix				
		Accuracy	Specificity	Sensitivity						
1.	Zainab Shafique	87%	81%	66%	F1=Aro, F2=Atom Q, F3=Atom Q, F4=hyd		F1	F2	F3	F4
						F1	0	5.50	6.0	3.27
						F2	5.50	0	7.58	6.49
						F3	6.0	7.58	0	2.89
						F4	3.27	6.49	2.89	0
2.	Aisha Arshad	87%	66%	100%	F1=Aro, F2=Aro, F3=Hyd		F1	F2	F3	
						F1	0	7.08	4.17	
						F2	7.08	0	3.27	
						F3	4.17	3.27	0	
3.	Maria Bibi	80%	85%	66%	F1=Acc F2=Acc F3=Aro F4=Hyd		F1	F2	F3	F4
						F1	0	2.6	2.79	7.54
						F2	2.6	0	4.97	2.54
						F3	2.79	4.97	0	4.97
						F4	7.54	2.54	4.97	0
4.	Ishrat	80%	87%	77%	F1=PiN F2=Hyd F3=Hyd F4=Aro Hyd		F1	F2	F3	F4
						F1	0	5.04	8.63	8.02
						F2	5.04	0	7.00	3.39
						F3	8.63	7.90	0	5.67
						F4	8.02	3.39	5.67	0
5.	Mahreen Nawaz	80%	83.3%	77%	F1=Hyd F2=Hyd F3=Acc2 F4=Don&Acc		F1	F2	F3	F4
						F1	0	3.50	7.79	9.15
						F2	3.50	0	4.88	6.17
						F3	7.79	4.88	0	4.44
						F4	9.15	6.17	4.44	0

Discussion:

Discussion of the all the developed models and advocate the finally selected model, in your discuss focus on the following points for each model,

1. Which compound/compounds were used to build the query?

Template with **low EC50 value** is used to build a query this is because a lower EC50 means the compound requires a lower concentration to achieve half of its maximum effect, indicating strong interaction with the target. In our project 5 models were build by selecting the same template with different features. The selecting value is most preferable because of its highest activity.

The template used in all 5 models is molecule 3 with EC50 value of 0.026 which is lowest among all molecules. The summary is given in table 2.

Table 2: summary of all the models

Models	Common feature	Remaining feature	TP/TN	FP/FN	Screening hits
1	hyd	Aro, AtomQ, AtomQ	9/4	2/0	2 molecule (5&8)
2		Aro, Aro	9/4	2/0	1 molecule (7)
3		Hyd, Acc, Acc, Aro	7/4	1/2	0
4		PiN, Aro hyd	7/5	1/2	0
5		Hyd, Acc2, Don&Acc	7/4	1/2	4 molecules (1,4,5&6)

2. Which features were used? In which distance to each other (use MOE/Edit/Measure/ Distances...) are the features? Include a picture.

The Final model is model number 1. Four features were selected in a template in which two of them were Atom Q with a radius of 0.6 these are the electronic clouds in a molecule, another one is an Aromatic of radius 0.5 which is also selected in other models such as model number 2, 3 and 4. The most common feature that was selected in all models is Hydrogen molecules in our model it has a radius of 0.4. Overall, hydrogen feature being prominent suggests that our models consider hydrogen interactions as a major determinant of biological or chemical behavior, which is valuable when refining compounds for desired bioactivity or drug-like properties.

The distance matrix of final model is given in table 3 and figure 1.

Table 3: Distance matrix of the final model

	F1	F2	F3	F4
F1	0	5.50	6.0	3.27
F2	5.50	0	7.58	6.49
F3	6.0	7.58	0	2.89
F4	3.27	6.49	2.89	0

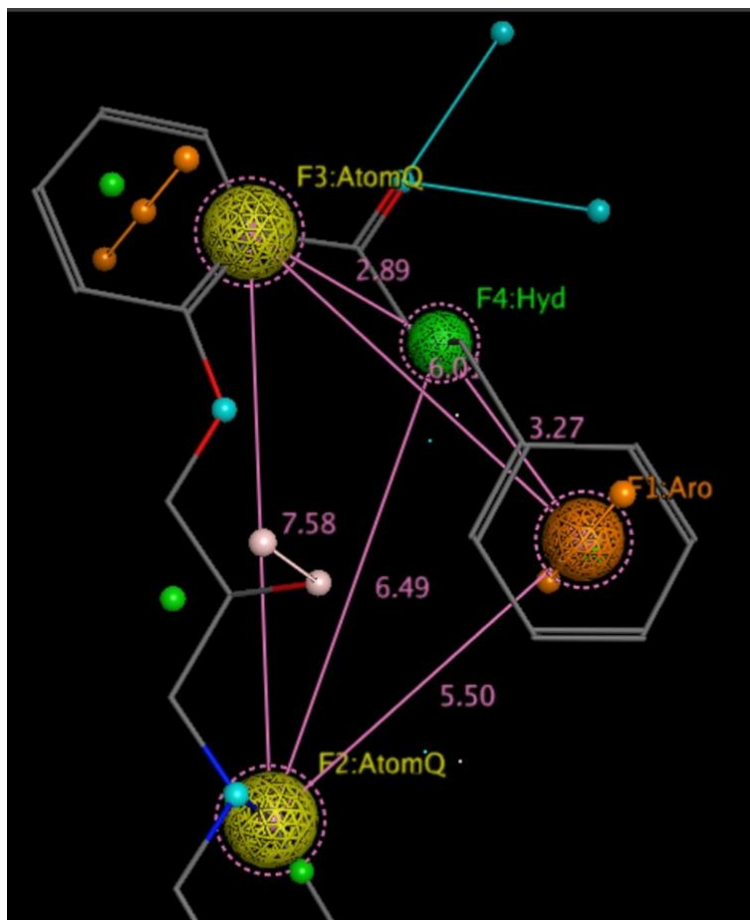


Figure 1: Distance matrix of the final model

3. Can the model distinguish between active and inactive compounds? Which compounds were identified, which not?

A threshold was established according to their EC50 values with values lower than 1 considered as actives while greater than 1 were considered as inactive. Yes, the model appears to effectively distinguish between active and inactive compounds based on their EC50 values. Active compounds are typically those with lower EC50 values (indicating higher potency), while inactive compounds have higher EC50 values (indicating lower potency or no significant activity).

From the table, we can observe that the compounds in the above 9 blue-highlighted rows (with EC50 values ranging from 0.0267 to 0.9073) are classified as active. These compounds have low EC50 values, suggesting strong activity against the target.

The compounds below the blue-highlighted section (with EC50 values ranging from 5.0600 to 82.6880) are classified as inactive, as they require much higher concentrations to achieve 50% of their maximal effect, indicating weaker or no significant activity.

File Edit Display Compute Window Help					SVL DBV MOE	⚙
	mol	mol:conf	Code	EC50		
1	3	3 -128	3	0.0267		
2	10	3 -128	10	0.1400		
3	5□	3 -128	5	0.1770		
4	9□	3 -128	9	0.2120		
5	1	3 -128	1	0.2800		
6	15	3 -128	15	0.3300		
7	6	3 -128	6	0.5330		
8	11	3 -128	11	0.8990		
9	2	3 -128	2	0.9073		
10	14	3 -128	14	5.0600		
11	7	3 -128	7	10.4670		
12	8□	3 124 2	8	11.1940		
13	4	3 -128	4	11.5170		
14	12	3 -128	12	11.6600		
15	13	3 -128	13	82.6880		

Figure 2: Hits selected by the final model in the training data set

The model correctly identified the first nine compounds (highlighted in blue) as hits (actives), as these have low EC50 values. The remaining compounds were correctly not identified as hits, as they have significantly higher EC50 values, corresponding to inactives. This suggests that the model is capable of distinguishing between active and inactive compounds based on EC50 thresholds as shown in figure 2.

4. Are similar compounds identified in a similar manner? Include some pictures of the identified conformations.

Yes, similar compounds appear to be identified in a similar manner based on the Root Mean Square Deviation (RMSD) values in the table on the right. RMSD is a measure of the similarity between molecular conformations, with lower RMSD values indicating higher similarity.

In the provided data shown in figure 3:

1. Low RMSD Values for Positive Hits:

- The blue-highlighted rows, which represent the positive hits, generally show low RMSD values. For example, compounds like 3 and 7 have RMSD values close to zero, indicating that these conformations closely match their reference structures or templates. This low RMSD suggests that these compounds have similar conformational features or orientations associated with activity, which the model recognizes as favorable.

2. Patterns of Conformational Similarity:

- Compounds such as 5, 6, and 7, which are highlighted in blue as active hits, show consistent low RMSD values, indicating structural alignment with the active template or model features. This implies that the model is identifying hits based on structural similarity, likely through the alignment of key functional groups or molecular shapes.

3. Consistency Across Models:

- Since these compounds were consistently flagged as positive across multiple models (as seen in previous data), it suggests that similar structural features are recognized by each model. This structural consistency provides further confidence that the identified conformations align with desired activity profiles, like hERG inhibition or binding affinity.

4. Differences in Higher RMSD Values:

- Compounds with higher RMSD values, like 12 (0.3980), are also active but may adopt different conformations than other identified hits. This might indicate some structural diversity among the hits, which could be useful for identifying structurally diverse compounds with similar activity.

1

File	Edit	Display	Compute	Window	Help	SVL
						mol
1	3	3-128				
2	10	3-128				
3	90	3-128				
4	90	3-128				
5	1	3-128				
6	15	3-128				
7	6	3-128				
8	11	3-128				
9	2	3-128				
10	14	3-128				
11	7	3-128				
12	40	3-124				
13	4	3-128				
14	12	3-128				
15	13	3-128				

						mol	mseq	rmsd
1	3					1	0.0000	
165	10					2	0.3149	
167	50					3	0.1431	
398	90					4	0.3097	
412	1					5	0.3615	
418	15					6	0.0266	
555	6					7	0.0104	
679	11					8	0.2101	
684	2					9	0.0000	
873	7					10	0.0505	
958	12					11	0.3988	

2

File	Edit	Display	Compute	Window	Help	SVL	DBV	MOE
						mol	molconf	Code
1	3	3-128	3					0.0207
2	10	3-128	10					0.1400
3	90	3-128	5					0.1770
4	90	3-128	9					0.2120
5	1	3-128	1					0.2800
6	15	3-128	15					0.3300
7	6	3-128	6					0.5330
8	11	3-128	11					0.8990
9	2	3-128	2					0.9873
10	14	3-128	14					5.0600
11	7	3-128	7					10.4670
12	40	3-124	8					11.1940
13	4	3-128	4					11.5170
14	12	3-128	12					11.6600
15	13	3-128	13					82.6880

						mol	mseq	rmsd
1	3					1	0.0000	
2	10					2	0.2182	
3	90					3	0.0827	
4	90					4	0.2171	
5	1					5	0.1447	
6	15					6	0.0000	
7	6					7	0.0000	
8	11					8	0.3839	
9	2					9	0.0000	
10	7					10	0.0000	
11	12					11	0.1447	

3

File	Edit	Display	Compute	Window	Help	SVL	DBV	MOE
						mol	molconf	Code
1	3	3-128						
2	10	3-128						
3	90	3-128						
4	90	3-128						
5	1	3-128						
6	15	3-128						
7	6	3-128						
8	11	3-128						
9	2	3-128						
10	14	3-128						
11	7	3-128						
12	40	3-124						
13	4	3-128						
14	12	3-128						
15	13	3-128						

						mol	mseq	rmsd
1	3					1	0.0000	
2	10					2	0.2294	
3	90					3	0.2186	
4	90					4	0.2462	
5	1					5	0.2982	
6	15					6	0.0000	
7	6					7	0.2093	
8	11					8	0.3216	
9	2					9	0.0000	
10	14					10	0.3823	
11	7					11	0.3890	
12	13					12	0.3147	

4

File	Edit	Display	Compute	Window	Help	SVL	DBV	MOE
						mol	molconf	Code
1	3	3-128						
2	10	3-128						
3	90	3-128						
4	90	3-128						
5	1	3-128						
6	15	3-128						
7	6	3-128						
8	11	3-128						
9	2	3-128						
10	14	3-128						
11	7	3-128						
12	40	3-124						
13	4	3-128						
14	12	3-128						
15	13	3-128						

						mol	mseq	rmsd
1	3					1	0.0410	
67	50					2	0.2625	
75	90					3	0.3737	
78	15					4	0.0701	
110	6					5	0.1409	
133	11					6	0.2658	
139	2					7	0.0707	
227	7					8	0.2619	

5

File	Edit	Display	Compute	Window	Help	SVL	DBV	MOE
						mol	molconf	Code
1	3	3-100	3					0.0207
2	10	3-100	10					0.1400
3	90	3-100	5					0.1770
4	90	3-100	9					0.2120
5	1	3-100	1					0.2800
6	15	3-100	15					0.3300
7	6	3-100	6					0.5330
8	11	3-100	11					0.8990
9	2	3-100	2					0.9873
10	14	3-100	14					5.0600
11	7	3-100	7					10.4670
12	40	3-100	8					11.1940
13	4	3-100	4					11.5170
14	12	3-100	12					11.6600
15	13	3-100	13					82.6880

						mol	mseq	rmsd
1	3					1	0.0000	
2	10					2	0.0000	
3	90					3	0.0000	
4	90					4	0.0000	
5	1					5	0.0000	
6	15					6	0.0000	
7	6					7	0.0000	
8	11					8	0.0000	
9	2					9	0.0000	
10	14					10	0.0000	
11	7					11	0.0000	
12	40					12	0.0000	
13	4					13	0.0000	
14	12					14	0.0000	
15	13					15	0.0000	

Figure 3: Comparison of all the models, its selected hits with their best confirmation. Figure 1 (Zainab Shafique), Figure 2 (Aisha Arshad), Figure 3 (Maria Bibi), figure 4 (Ishrat), figure 5 (Mahreen Nawaz).

5. Explain the predicted behaviour of the compounds in the unknown data.

In predicting the behavior of compounds in unknown data, the model's performance is inferred from how it identified active hits during screening in the known data set. Given that the model identified varying numbers of hits across five screening models, we can analyze this to understand its behavior on unseen data. Here's how each model's performance can be interpreted based on the data provided:

1. Model with 2 Hits (Molecules 5 & 8):

- This model identified two active compounds. This suggests it may have focused on certain structural or functional features shared by compounds 5 and 8. If these features are present in new compounds, this model is likely to predict them as active. However, it could be more restrictive, capturing a narrower range of active compounds.

2. Model with 1 Hit (Molecule 7):

- Identifying only one active compound implies that this model has a stricter or more specific criterion for activity. It may be looking for unique features that make molecule 7 stand out from the others. This model might miss compounds with moderate activity but could accurately predict highly potent actives in unknown data.

3. Models with 0 Hits:

- The two models with no hits may be too stringent in their criteria or are possibly optimized for different features that aren't as relevant for identifying activity in this dataset. These models may struggle with sensitivity and could miss active compounds in unknown data unless they match specific, possibly non-representative, structural traits.

4. Model with 4 Hits (Molecules 1, 4, 5 & 6):

- This model shows the highest sensitivity by identifying four actives, including compounds with moderately low EC50 values. This suggests that the model has a broader or more flexible set of criteria, allowing it to capture a wider range of active compounds. It might perform well in identifying hits in unknown data, especially compounds with varying potencies.

Final model:

The first model is chosen because it provides a higher accuracy of 87%, ensuring that the hits it identifies are more likely to be true positives. This balance between accuracy and a reasonable number of hits makes it more reliable for screening unknown data, reducing the risk of false positives and optimizing the follow-up testing process. In screening contexts, especially early drug discovery, accuracy and precision are often more valuable than sheer sensitivity, making the first model the optimal choice.

Molecule 5 has been identified as a potential hERG inhibitor by two separate models, including the final chosen model, suggesting it has structural features associated with hERG activity. This consistent identification strengthens confidence in its activity. Its low EC50 value (0.1770) indicates high potency, which can be both advantageous and concerning. While potent inhibitors are desirable for efficacy, strong hERG inhibition raises the risk of cardiotoxicity, a critical safety concern in drug development.